



电子科技大学
格拉斯哥学院
Glasgow College, UESTC

Embedded Processors UESTC2004 – 2024-25

Semester 2

Written Assignment

Smart Locker Security and Environmental Monitoring System

Student's Chinese name	宋泽旭
Student's English name	Song Zexu
Student's UESTC ID#	2023190902025
Student's UoG ID#	2960231S

Content

Introduction-----1

Methodology-----2

- 1. Material required-----2
- 2. Connection-----2
- 3. Code-----3

Result and discussion-----4

Conclusions and Future Work-----5

References-----5

Appendices-----6

Introduction

This project is required to build a Smart Locker Security and Environmental Monitoring System. A Nucleo L432KC mbed board is the most important hardware of this project. Besides, TMP102 is required to estimate the temperature, and ADXL345 is required to estimate the acceleration. A buzzer, two LED lights, two buttons, and wires are required.

The system includes security features for user authentication. Users must enter a password via the buttons. If the password is incorrect, the red LED will turn on and the buzzer works as a warning. When the correct password is entered, the green LED lights up.

In addition to password verification, environmental monitoring functions are integrated using the TMP102 temperature sensor and ADXL345 accelerometer. This report will detail the hardware setup, password authentication logic, and sensor integration, ensuring reliable performance and system security.

Methodology

1. Material required

In this project, a Nucleo L432KC mbed board served as the main controller for managing system operations, including password verification. To monitor environmental conditions, a TMP102 temperature sensor was used to provide accurate temperature readings, while an ADXL345 accelerometer was implemented to measure acceleration data. A green LED was used to indicate a successful password entry, whereas a red LED was employed to alert the user of incorrect password attempts. Two push buttons were used. To provide feedback to the user, a buzzer is used to make sound. Finally, additional wires were used to establish reliable connections between all the hardware components and the mbed board.

2. Connection

In this project, I2C and SPI connection were both used to promise the connection between master and slave. Firstly, the buzzer was connected to D13 for warning. LED were connected to D4, D5 pins. The buttons were connected to D9 and D10. The most importantly, TMP102 was connected as SDA to D12, SCL to A6. As a result, TMP102 were used I2C to connected with L432KC. In addition, ADXL was also using I2C to connect. Here is an actual circuit connection.

Specific connection would be shown in Appendix A

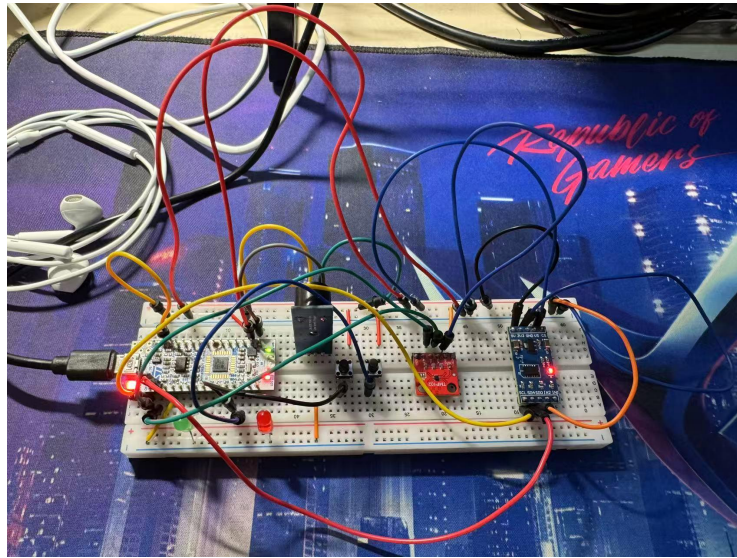


FIG1. Actual Circuit Connection

3. Code

The code is written as in a modular approach, to make main function easier to read and understand. Specific code is in Appendix B.

Result and discussion

Not only using buzzer and LED to show when the environment condition changes, it is also possible to see the specific numbers in PC.

```
X: 0.05 g, Y: -0.03 g, Z: 0.92 g
Temperature in Celsius= 27.188
Temperature in Celsius= 27.188
X: 0.05 g, Y: -0.02 g, Z: 0.92 g
Temperature in Celsius= 27.188
Temperature in Celsius= 27.125
X: 0.05 g, Y: -0.02 g, Z: 0.92 g
Temperature in Celsius= 27.125
Temperature in Celsius= 27.125
X: 0.05 g, Y: -0.02 g, Z: 0.92 g
Temperature in Celsius= 27.125
Temperature in Celsius= 27.063
X: 0.05 g, Y: -0.03 g, Z: 0.92 g
Temperature in Celsius= 27.063
█
```

FIG2.Specific numbers

In the end, this project realized a Smart Locker Security and Environmental Monitoring System. For the code part, if user enter a wrong code, red light would turn on until user enter the wright code.

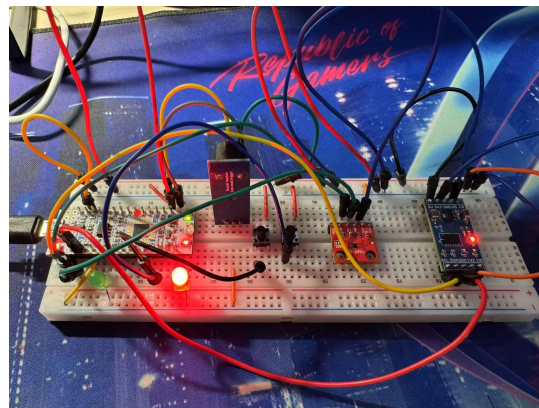


FIG3. Wrong Code Enter

For the temperature monitoring part, when the temperature exceeds the preset 29 degrees Celsius, buzzer will continuously sound the alarm.

For the movement part, If the acceleration change is more than 1.2g, the alarm will be triggered.

After such testing, it is believed that the system could work and temperature and movement could be tested. Also, the system could make correct judgments and provide corresponding feedback.

Conclusions and Future Work

In this project, a varies of problems appeared. As online platform cannot be used to compile, off-line MBED STUDIO has been used. However, Perhaps limited by the region, the main function library of mbed cannot be called correctly. Therefore, I have chosen to download all the function libraries in my own project folder.

Apart from such software issues, this project cannot meet the standard of lower power mode. Also, only use LED and buzzer may not show it clearly about the feedback after user make inputs. Thus, an OLED screen or a segment display may be a better choice than this time.

In the future work, the process of doing the entire project can be optimized. First, plan all the materials required for the project and the effects these materials will produce. Then, select those that best meet the project requirements and are most convenient for users to operate, to avoid such phenomena.

References

- [1] R. Brzoza-Woch, Ł. Gurdek and T. Szydło, "Rapid Embedded Systems Prototyping - an Effective Approach to Embedded Systems Development," 2018 Federated Conference on Computer Science and Information Systems (FedCSIS), 2018, pp. 629-636.
- [2] R. Toulson and T. Wilmshurst, Fast and Effective Embedded Systems Design: Applying the ARM Mbed, Oxford, UK: Elsevier, 2012.

Appendices

Appendix A

Pin	Macro Name	Object Type
D1	ADXL SCL	I2C
D1	TMP SCL	I2C
D0	ADXL SDA	I2C
D0	TMP SDA	I2C
D4	Green LED	
D9	Button 1	
D10	Button 2	
D12	Red LED	
D13	Buzzer	

[Appendix B](#)

In Folder *Appendix B Code*